CCS 2501 – Case Study #2, QuickBiz Messenger

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CHAPTER 1

Introduction

1.1 Background of the Company

QuickBiz Messenger is a delivery service firm that deliver packages and was established by Andrew Langston who obtained the idea after working for a local bicycle delivery service during his college years. After returning home, he set up QuickBiz Messenger as a business in his hometown Seattle which at that time only had a handful of small messenger services so there was not that much competition. In its early years, Andrew Langston worked alone and wrote every log sheet and customer slip by hand. QuickBiz Messenger used Bicycle as a means of service when delivering messages and packages to its clients. Opportunity came for QuickBiz Messenger in the 1990’s when the pace of business transactions skyrocketed and firms of all sizes needed additional services to carry out their day-to-day transactions so the number of deliveries required also increased. Traffic on the metropolitan area of Seattle also became heavier over time so delays were common for most of the other messenger services. With this situation, QuickBiz Messenger and through its model of using bicycle as a means to deliver which did not have much problems with the traffic meant that packages were always delivered safe and on time which allowed QuickBiz Messenger to build its image as a reliable service with good reputation.

1.2 Current System

As business became bigger and more complex, Andrew hired Sarah Truesdale to be his bookkeeper and receptionist. Sarah organized an office and set up basic business applications such as word processing, spreadsheet, and database programs on the company’s first PC. When Andrew received a delivery request, Sarah typed it in her daily log sheet and Andrew went to cycle and deliver. As the customer base grew, Andrew began hiring college students as part-time biker messengers and would be given instructions by dispatchers on new orders and delivery requests through the SMS. QuickBiz Messenger currently has more than 90 employees. Deliveries were kept track through No-Carbon-Required (NCR) Forms which were held by the couriers and would give them to the customers to fill out delivery information. Customers kept a copy of the form, and the couriers took the originals back to the main office. Sarah would then input the customer information such as order number, address, and type of service along with the facts of the delivery such as start and end times, courier name, and delivery address. Andrew recently expanded his services to car and truck deliveries which meant management of the messengers became more complex as these services do not work well with traffic which the company was known for not having problems with. Andrew is also planning to be able to assess the performance of his employees by tracking each of the messenger's number of customer compliments and complaints, number of deliveries, and their shortest delivery times. He also plans to be able to evaluate the different delivery territories to determine which were the most profitable and to generate sales reports by region.

1.3 Problems of the Current System

QuickBiz Messenger currently has 5 problems:

(1) Tracking the availability of the Employees

- The growth of messengers in the company is causing problems for Sarah Truesdale and Leslie Chen in manually managing the schedule for the messengers which became more complicated overtime. It should also be mentioned that most of the employees that were hired were part-time workers so certain individuals are only available while others are not in certain times. Most of these are College Students so their schedule would change every semester. Additionally, when some were suddenly unavailable in the moment such as calling-in sick or having emergencies, the managers would scramble in lining up a replacement which were available for the delivery.

(2) No existing methods to assess the performance of the Employees

- Andrew wants to know which of the messengers in his company perform the best for them to receive a bonus from outside feedback and especially customers. As of the moment however, QuickBiz has no available method for it.

(3) No existing methods to evaluate the Territories

- Andrew wants to have an idea on which of the territories they operate in are the most profitable but as of the moment, QuickBiz has no available method for it.

(4) Delayed Deliveries due to some Messengers being unfamiliar with the routes

- Most of the long-time messengers in QuickBiz Messenger already know the most efficient routes in Seattle with the exception for some of the few recently employed messengers who were not native to the area. This caused delays with some of the delivery. Customers and clients affected by this are beginning to complain and QuickBiz Messenger recognizes this problem as a threat to their reputation.

(5) Outdated Tracking Delivery Data

- Problems arose for QuickBiz Messenger in using the NCR Forms as a method to record delivery information when some of the handwritings were difficult to read and would often be damaged due to the weather. Because of this, inputting the data became tedious and was even worsened after the company expanded its services to cars and trucks. The number of NCRs required to review by Sarah became more unmanageable over time.

Making an analysis, this identifies two possible root causes:

(1) The company’s information management is limited and only tracks the delivery data which meant that having useful data and information about their employees and the territories they operate in cannot be managed, automated, nor observed accurately. This caused problems in their management and data gathering such as having difficulty in assigning schedules with their employees, not being able to easily assess the performance of their employees nor evaluate the profits of their territories.

(2) The company is not up-to-date with Information Technology such as still using NCR Forms for tracking delivery data and having no methods to automate some of their processes meant that with the ever-growing complexity of their operations, it is only a matter of time that they would find problems that those who are up-to-date with Information Technology do not have.

1.4 Objectives of the System

General Objective

- To solve the 5 main problems of the current system that was observed by QuickBiz Messenger using a new and improved system that the group will be developing.

Specific Objectives

Redesigning and improving the existing Database

- The database must not only be able to store data details of the deliveries that were undertaken by QuickBiz Messenger, but also details about the employees of QuickBiz Messenger, details regarding the customers that are using the services of QuickBiz

Messenger, and data regarding the different territories the company operates in. This objective is important as this will be the first objective that will introduce the company in automation and exploring digital processes.

Designing and developing an Employee Management Program

- This program must be able to easily input and modify details about the employees including their schedules and availability in the database. This objective is also important as it will allow the managers to add and modify the data of the employees in the database which should offer a solution to the first root cause.

Designing and developing an Available Employees Viewer

- This program should be able to produce a list of the available employees based on the conditions inputted. This objective should be able to offer a solution in challenges with Tracking the Availability of the Employees.

Designing and developing an Employees Assessment

- This program should be able to list and assess the performance of the employees based on who has the shortest delivery time, has the most compliment, has the least complaints, and has the most deliveries. This program should be able to introduce a method that allows Andrew Langston to assess the performance of the employees.

Designing and developing a Territories Evaluation

- This program must be able to list and assess which of all the different territories has the most profits. This program should be able to introduce a method which will allow Andrew Langston to assess the performance of his employees and evaluate the total profit of the different territories.

Designing and developing a Path-Finder

- This program must be able to recommend and suggest paths that are efficient and used by the couriers that are familiar of the areas in Seattle. This is to be used by the couriers in their own device since they also receive orders through SMS. This objective aims to resolve the problem regarding some of the deliveries being delayed because of recently employed messengers being unfamiliar with the area.

Designing and developing an Application for the Customers

- This app must be able to register customers as official clients in order for them to use the services offered by QuickBiz Messenger such as being able to fill-in the delivery form in their own device and have an overview list of the delivery services that were offered to them. This objective aims to resolve the problem regarding the Outdated Tracking Delivery Data issue and replace the NCR Forms by digitizing and automating the processes.

1.5 Scope and Limitation of the System

This system will be focusing on giving broad and general solutions based only on all of the five problems that was recognized by the group. The group will only be limited to developing a Windows Form App (.NET Framework) using Visual Studio with a free MySQL Database Host to implement all of the possible solutions regarding the case. Most of the solutions in the system may also require some improvements especially regarding security, interface, and ease of use.

1.6 Significance of the System

The system to be designed and developed will allow QuickBiz Messenger to hopefully offer solutions in the 5 problems that were recognized and more importantly, digitize most of the company's operations and methods especially regarding data and information which will allow it to catch up with technological progress and trendings as the company thrives in a digital age where most things are automated.

CHAPTER 2

System Analysis and Requirements Modeling

2.1 Users of the System

Managers

- The managers include individuals such as Andrew Langston and Sarah Truesdale who will be using the Employee Management where they can add and update details regarding the employees working in the company.
- Managers specifically Sarah Truesdale and Leslie Chen will be using the Available Employees Viewer where they can input details such as Time Range Start, Time Range End, and which day. Afterwards, the system will then automatically list all of the employees that are available who match the inputted criteria from the database that meets the requirements of the entered settings.
- Manager Andrew Langston specifically will be using the Employees Assessment where he can sort a list of his employees in the company based on who has the shortest delivery time, has the most compliment, has the least complaints, and has the most deliveries. It should be noted that since this is determined every end of the month, the mentioned details regarding the employees must be reset to its default count which is 0.
- Manager Andrew Langston specifically will also be using the Territories Evaluation Program where he can sort a list of the territories QuickBiz Messenger operates in based on which are the most profitable.

Employees

- The employees are all of the messengers of the QuickBiz Messenger though it is likely that only the unfamiliar messengers would use the Path-Finder. In this program, employees can set the location that they are currently in and the location that they are heading towards. With these settings, the program can then suggest paths that are efficient and used by those familiar of the areas in Seattle.

Unregistered Customers

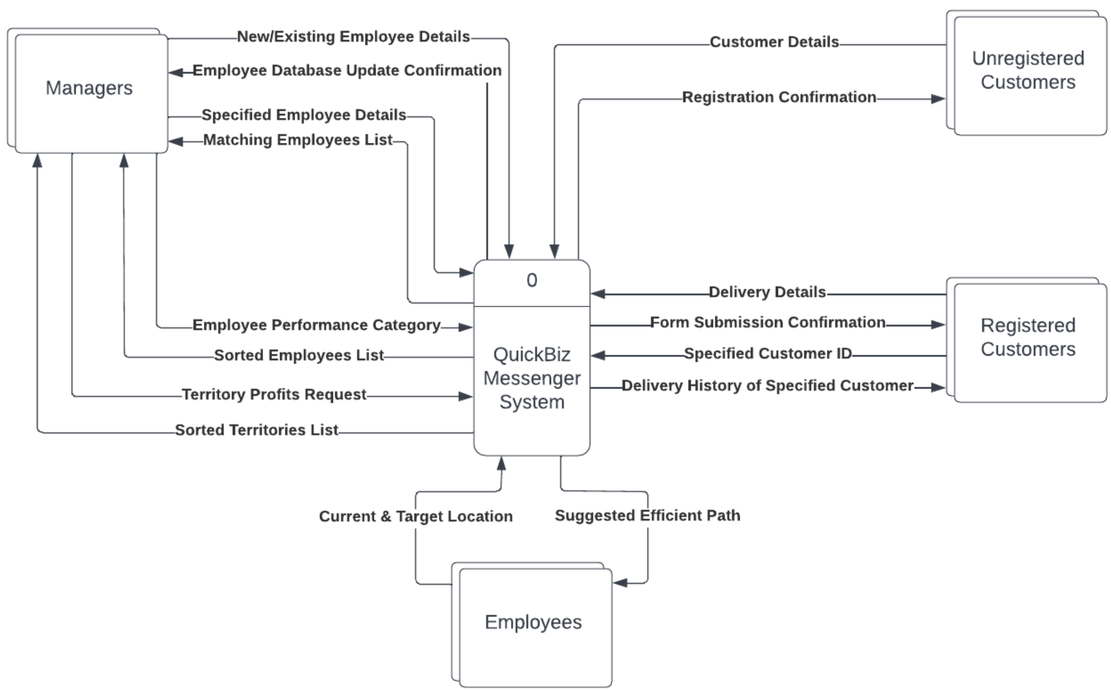
- The Unregistered Customers are the potential clients of QuickBiz Messenger. They will be using an application in their own device where they can register for an account before they can use the services offered by QuickBiz Messenger.

Registered Customers

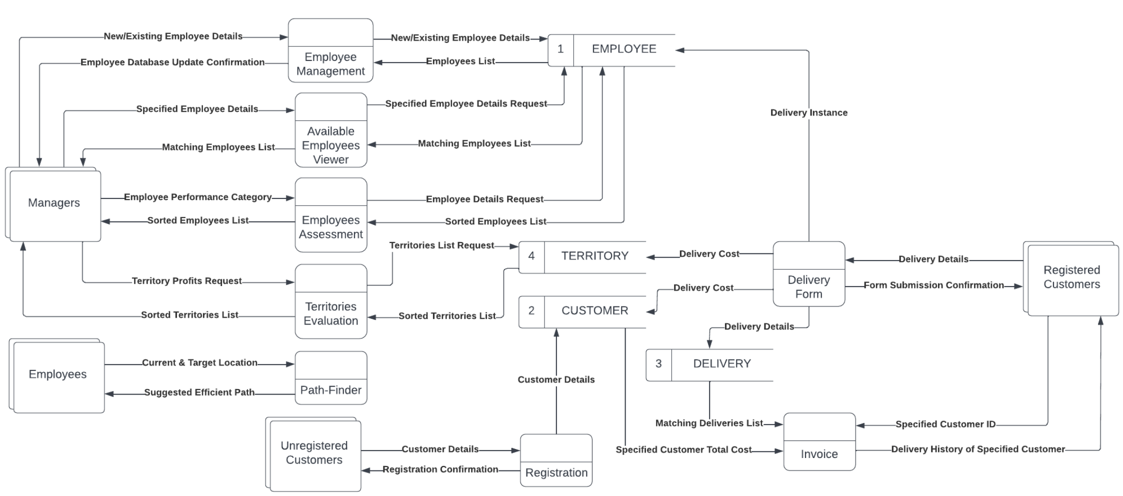
- The Registered Customers are the official clients of QuickBiz Messenger that can use the services offered by QuickBiz Messenger. A registered customer will be able to fill-in the Delivery Form which contains Delivery Information from the application that sends it to the database. A registered customer will also be able to have an overview of all the deliveries that they have requested from the application which replaces and acts as their monthly hard-copy invoice.

2.2 Proposed Data Flow Diagram

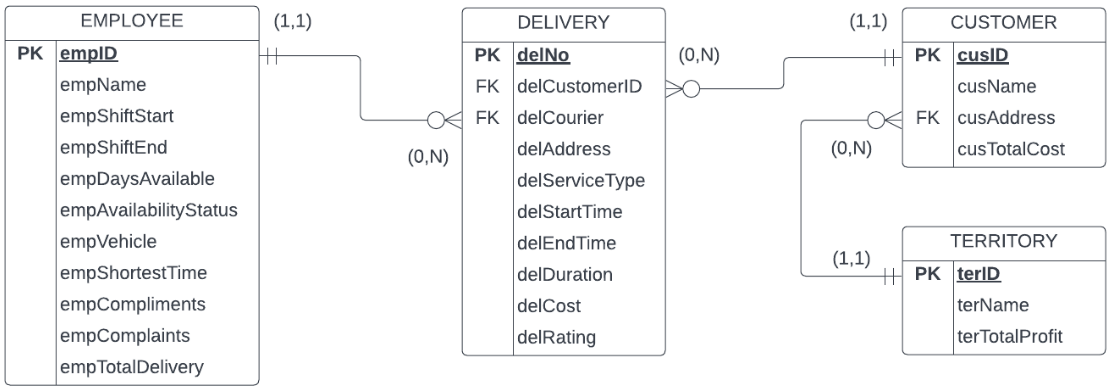
Context Diagram | QuickBiz Messenger



Level 0 Diagram | QuickBiz Messenger



2.3 Entity Relationship Diagram



2.4 Database Structure

Table Number: 1

Table Name: EMPLOYEE

Primary Key: empID

Foreign Key: NA

Table Description: This table is used to store an Employee's data. This is to be used for the Employees Management, Available Employees Viewer, Employees Assessment, and Delivery Form.

Field	Attributes	Data Type	Size	Content/Description
1	empID	Int	4	Employee's ID Number
2	empName	Varchar	30	Full Name of Employee
3	empShiftStart	Int	4	Starting Time of Employee's Shift
4	empShiftEnd	Int	4	Ending Time of Employee's Shift
5	empDaysAvailable	Varchar	30	Days Employee is Available in
6	empAvailabilityStatus	Varchar	10	Employee's Availability Status
7	empVehicle	Varchar	30	Vehicle of the Employee
8	empShortestTime	Int	4	Shortest Delivery Time of Employee
9	empCompliments	Int	4	Employee's Number of Compliments
10	empComplaints	Int	4	Employee's Number of Complaints
11	empTotalDelivery	Int	4	Employee's Number of Total Delivery

Table Name: CUSTOMER

Table Number: 2

Primary Key: cusID

Foreign Key: cusAddress

Table Description: This table is used to store a Registered Customer's data. This is to be used for the Registration, Delivery Form, and Invoice.

Field	Attributes	Data Type	Size	Content/Description
1	cusID	Int	4	Customer ID Number
2	cusName	Varchar	30	Full Name of Customer
3	cusAddress	Int	4	Territory ID of Customer's Address
4	cusTotalCost	Double	7,2	Total Cost of Customer from Deliveries

Table Name: DELIVERY

Table Number: 3

Primary Key: delID

Foreign Key: delCustomerAddress, delCourier

Table Description: This table is used to store a Delivery. This is to be used for the Delivery Form and Invoice.

Field	Attributes	Data Type	Size	Content/Description
1	delNo	Int	4	Delivery Number
2	delCustomerID	Int	4	ID of Customer in the Delivery
3	delCourier	Varchar	30	ID of the Employee in the Delivery
4	delAddress	Int	4	Address of the Customer in the Delivery
5	delServiceType	Varchar	30	Service Type of Delivery (Message/Package)
6	delStartTime	Int	4	Time that Delivery was Requested
7	delEndTime	Int	4	Time that Delivery was Received
8	delDuration	Int	4	Duration of the Delivery
9	delCost	Double	7,2	Cost of the Delivery
10	delRating	Varchar	10	Rating of the Employee in the Delivery

Table Name: TERRITORY

Table Number: 4

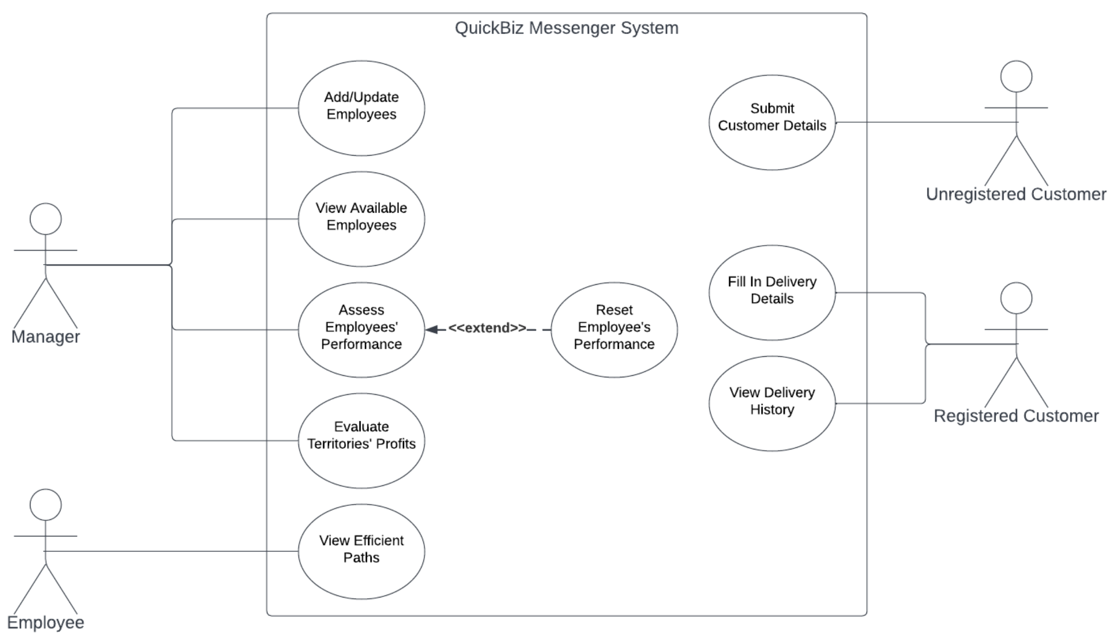
Primary Key: terID

Foreign Key: NA

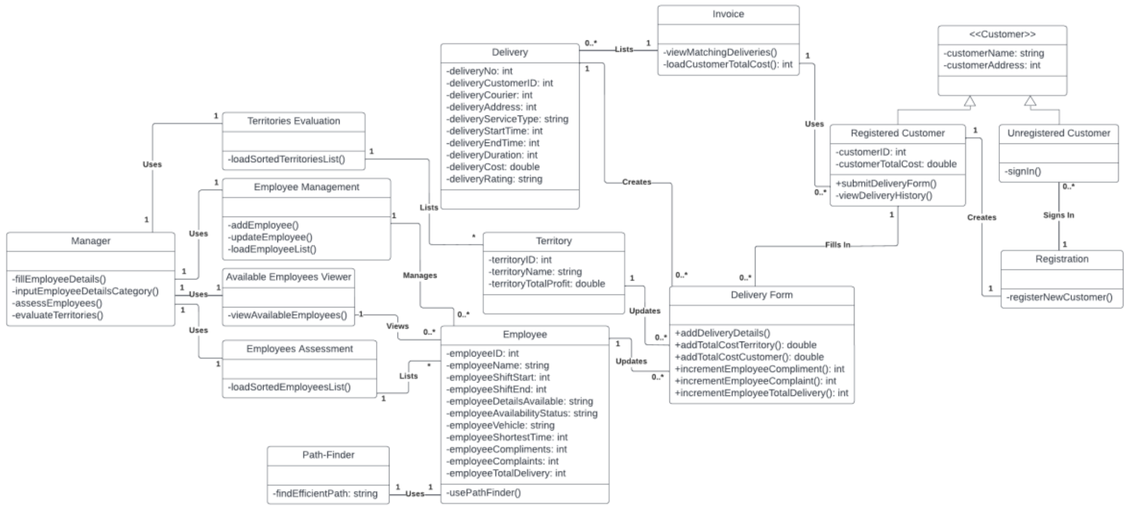
Table Description: This table is used to store a territory data. This is to be used for the Territories Evaluation, Registration, and Delivery Form.

Field	Attributes	Data Type	Size	Content/Description
1	terID	Int	4	Territory ID Number
2	terName	Varchar	10	Name of the Territory
3	terTotalProfit	Double	7,2	Total Profit of the Territory

2.5 Use-case Diagram



2.6 Class Diagram



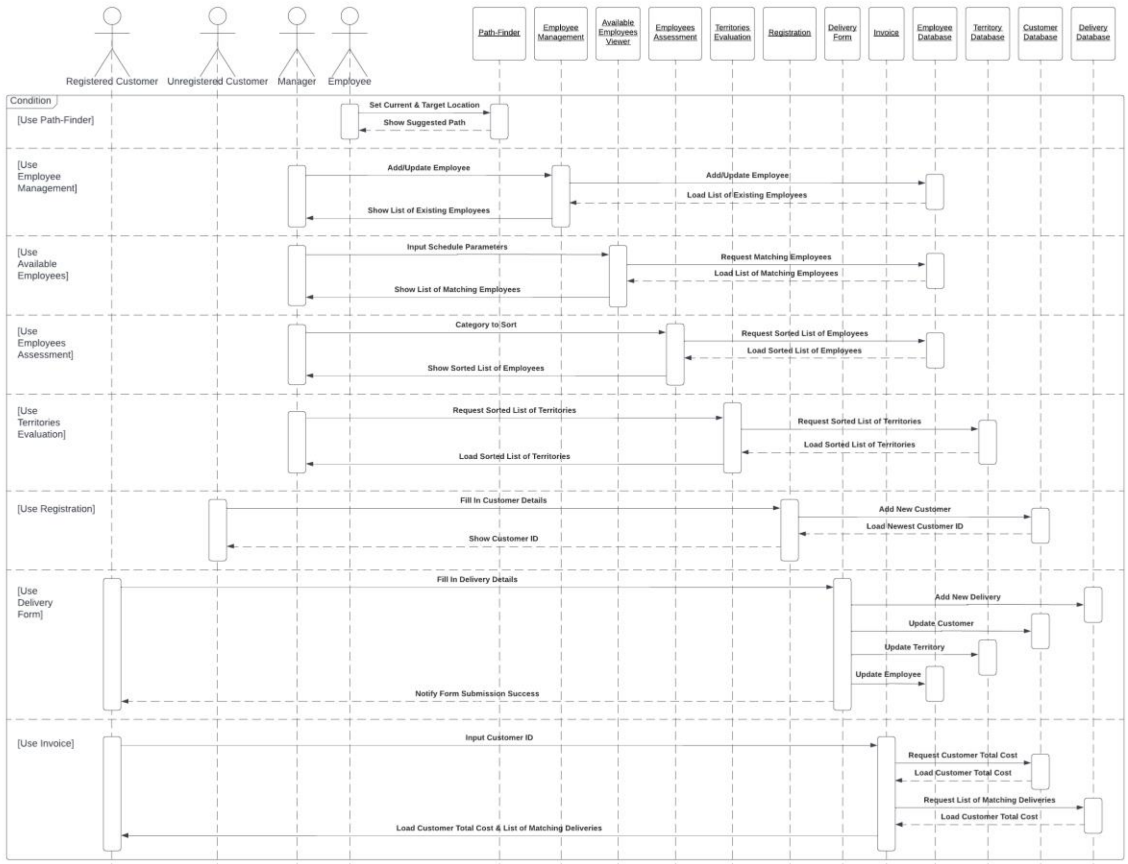
An Employee can only use Path-Finder which returns a suggested path for the Employee.

A Manager can use Territories Evaluation, Employee Management, Available Employees Viewer, and Employees Assessment. Territories Evaluation lists all of the Territories. Employee Management manages none or all of the Employees. Available Employees Viewer views which of the Employees have matching details and may return none or many Employees. Employees Assessment lists and sorts all of the Employees.

An Unregistered Customer can only use Registration which creates one and only one new Registered Customer.

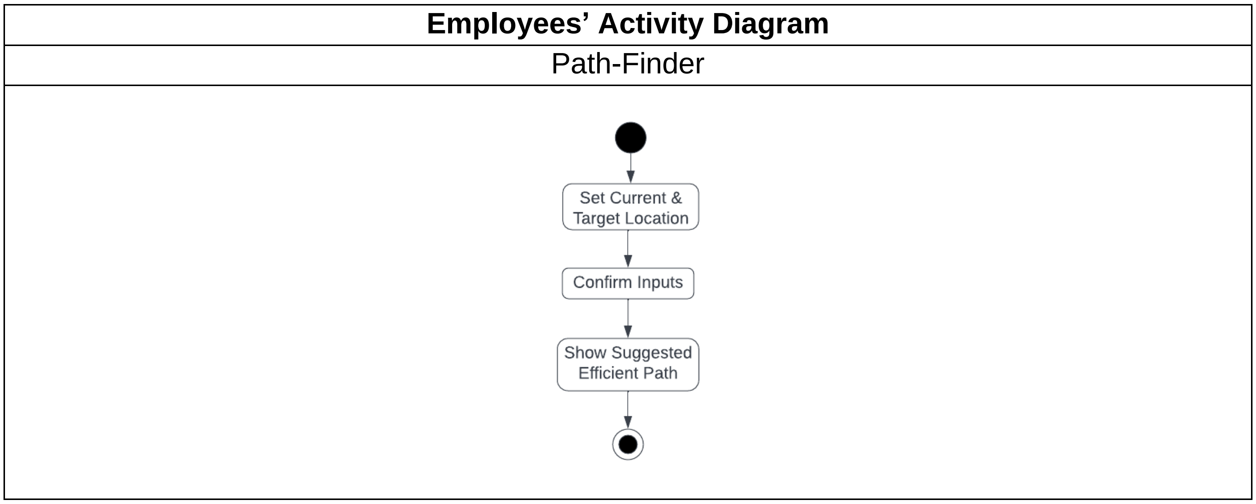
A Customer can use Delivery Form and Invoice. Delivery Form creates one and only one new Delivery, updates a Registered Customer, a Territory, and an Employee. Invoice lists which of the Deliveries have matching details and may return none or many Deliveries while also showing the Total Cost of the Customer.

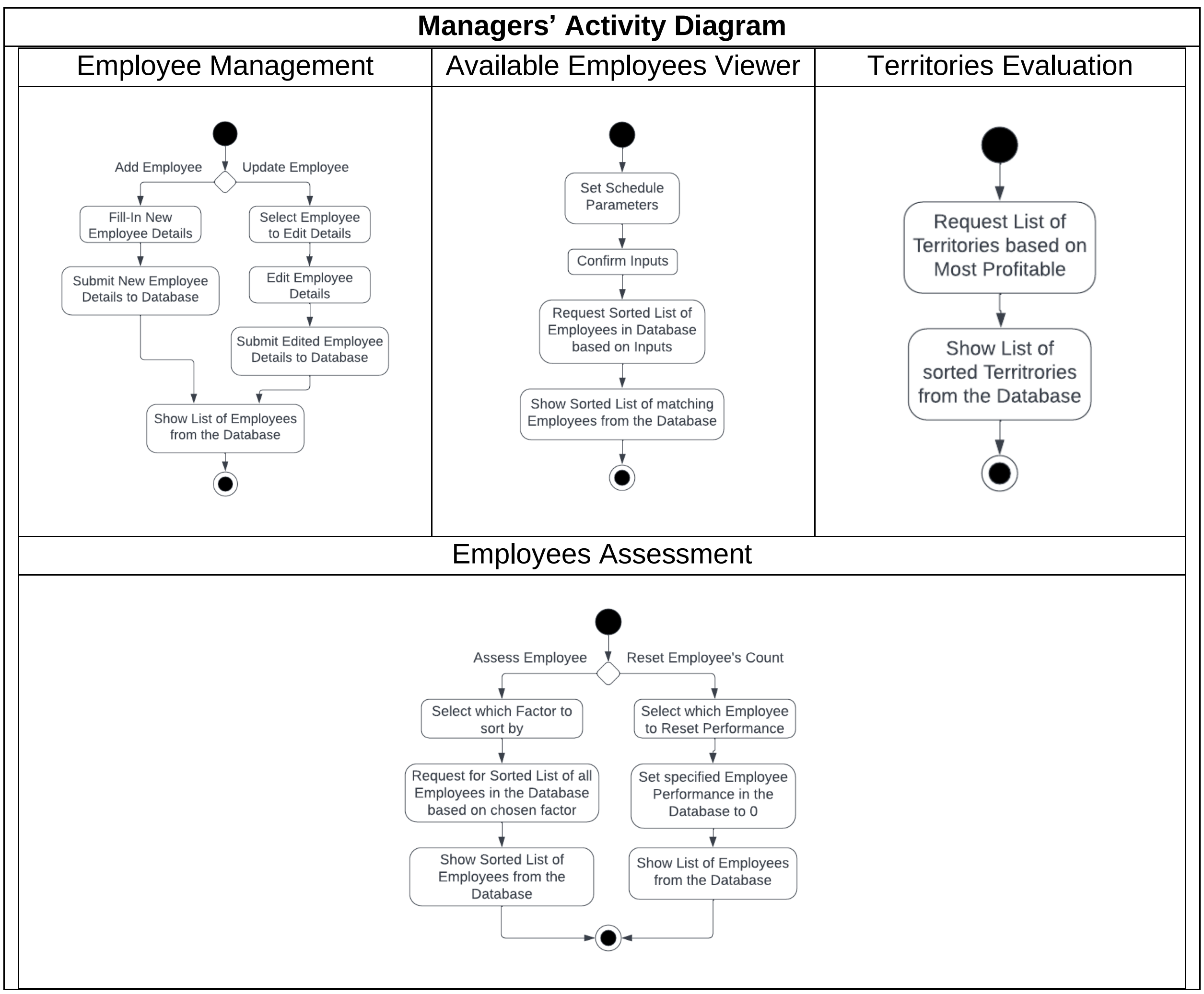
2.7 Sequence Diagram

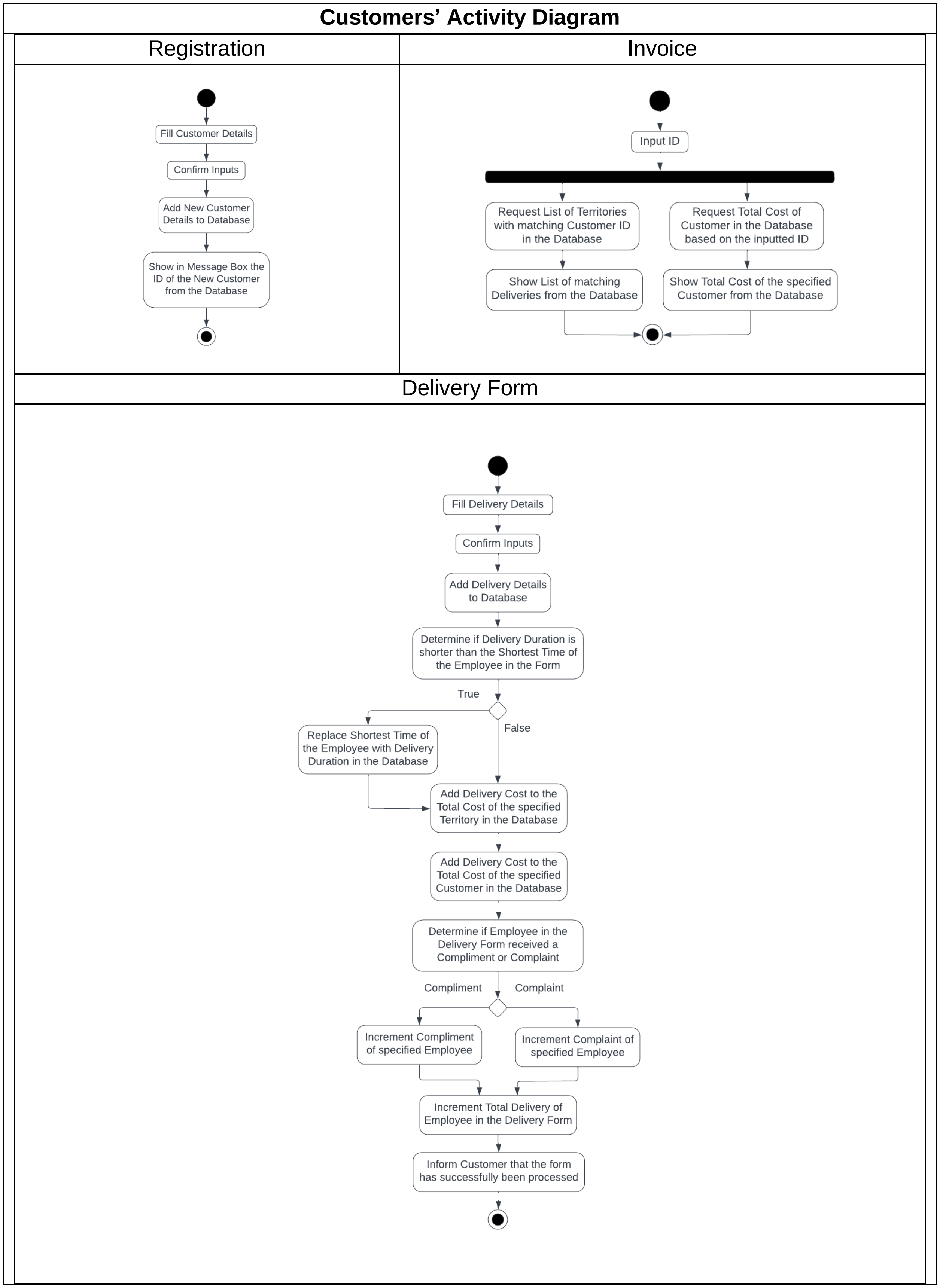


The Sequence Diagram above details the division about how Actors interact with the different objects. Note that the sequence to be focused more on in the system is the sequences with the specific objects that are divided as shown above since any of these grouped object sequence can be placed anywhere in the lifeline.

2.8 Activity Diagram






Customers' Activity Diagram
Registration
Fill Customer Details
Confirm Inputs
Add New Customer Details to Database
Show in Message Box the ID of the New Customer from the Database
Invoice
Input ID
Request List of Territories with matching Customer ID in the Database
Request Total Cost of Customer in the Database based on the inputted ID
Show List of matching Deliveries from the Database
Show Total Cost of the specified Customer from the Database
Delivery Form
Fill Delivery Details
Confirm Inputs
Add Delivery Details to Database
Determine if Delivery Duration is shorter than the Shortest Time of the Employee in the Form
True
False
Replace Shortest Time of the Employee with Delivery Duration in the Database
Add Delivery Cost to the Total Cost of the specified Territory in the Database
Add Delivery Cost to the Total Cost of the specified Customer in the Database
Determine if Employee in the Delivery Form received a Compliment or Complaint
Compliment
Complaint
Increment Compliment of specified Employee
Increment Complaint of specified Employee
Increment Total Delivery of Employee in the Delivery Form
Inform Customer that the form has successfully been processed

CHAPTER 3

Methodology

The group will be using the Waterfall Model as it is the most simple and straightforward approach.

Planning

In the planning phase, the group decided to review the case that was assigned then had a discussion on what was the background of the company, how it currently operated, identifying the requests and plans of the company, and what are the problems that the group could find. The group then discussed some plans and ideas on what could be the potential design of the system that will be used in the case.

Design

In the design phase, the group began suggesting and organizing ideas on what are some of the basic programs that could be done with the problems identified such as writing pseudocodes. This was continuously done until the group came up with an idea of a full and feasible system that has all of the programs and features that would act as the solutions to the recognized problems. The group then began working with the early diagrams as a model to base the system on.

Development

In the development phase, the group began with the programming of the system using Visual Studio to create a Windows Form Application. The group then created a MySQL Database Host as soon as most of the Windows Form Application's features were ready.

Testing

In the testing phase, the group began testing the prototype and would identify the errors and some of the features that needs improvement. This was done continuously until the system based on the design functioned without error and each of the programs could perform the tasks that it was designed to accomplish.

Deployment

In the deployment phase, the group decided that the system was functional and was ready to be used for the case. The group documented most of the system then deployed the system into the business operations of the case.

Maintenance

In the maintenance phase, the system will be continuously monitored for its performance, possible issues, and current state by the users of the system. The system will be updated should it be deemed necessary and this phase will continue until the users of the system has decided that the system is not viable anymore for the case.

CHAPTER 4

Budget and Proposal

Hardware Cost

Equipment	Cost
Local Database Server	54,000
Desktop Computer	21,000
24V Uninterruptable Power Supply	25,000
Ethernet Switch	2,000
Total	102,000

Software

	Cost
Programmer	24,000
User Interface Designer	38,000
System Administrator	35,000
Business Analyst / Economist	40,000
Total	137,000

Hardware Depreciation

Hardware Cost = 102,000   Depreciation Year = 5 Years

Rate of Depreciation = 20,400

Year	Amount
0	102,000
1	81,600
2	61,200
3	40,800
4	20,400
5	0

Software Deprecation Rate

Software Cost = 137,000   Depreciation Year = 10 Years

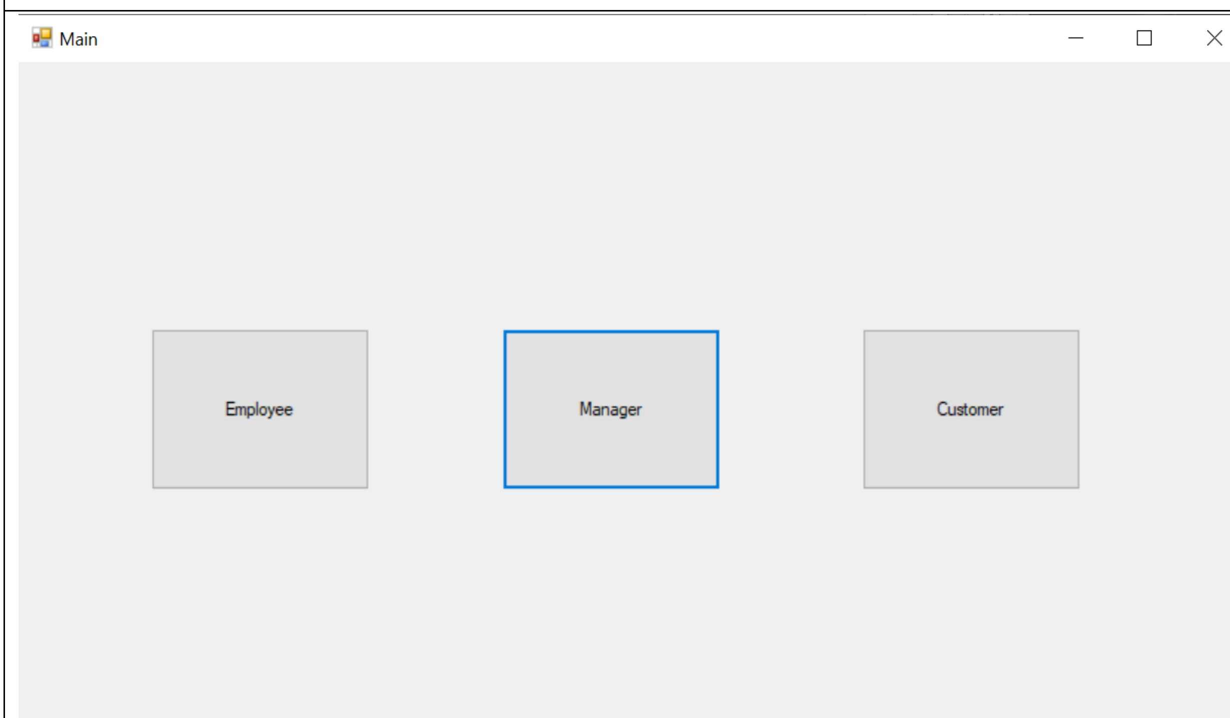
Rate of Depreciation = 13,700

Year	Amount
0	137,000
1	123,300
2	109,600
3	95,900
4	82,200
5	68,500
6	54,800
7	41,110
8	27,400
9	13,700
10	0

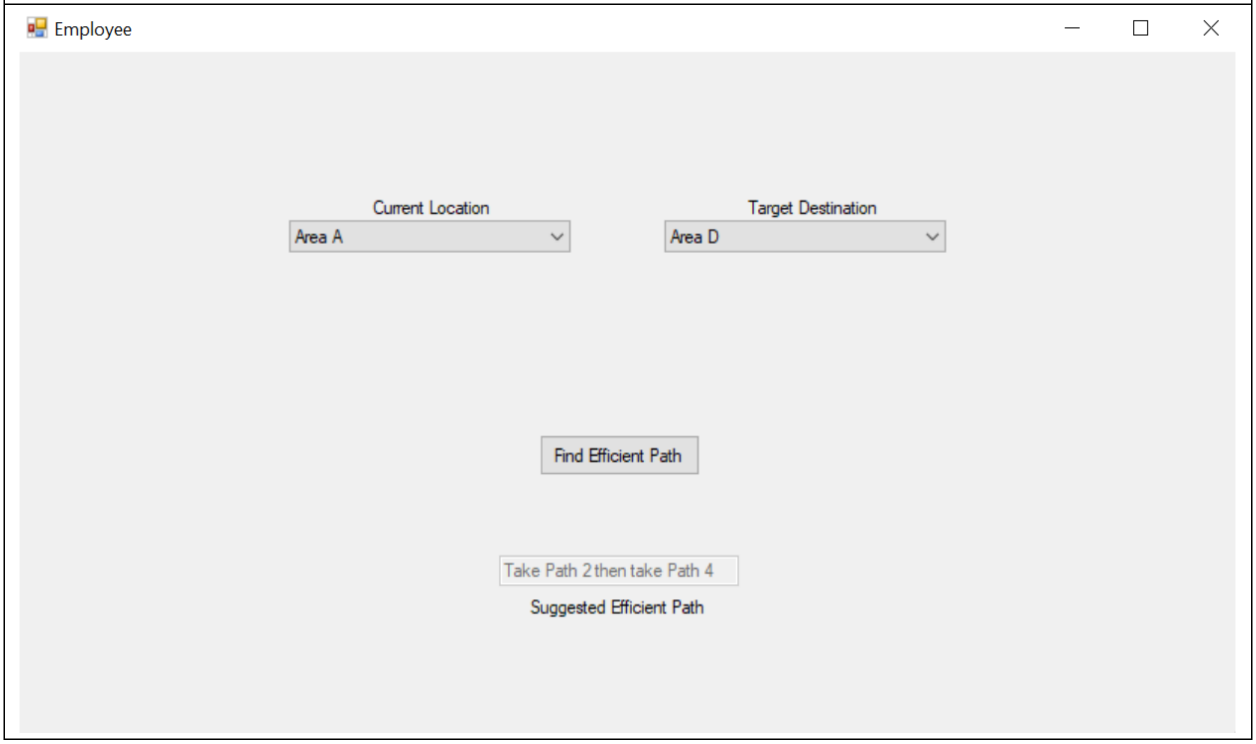
APPENDICES

Screen Shot

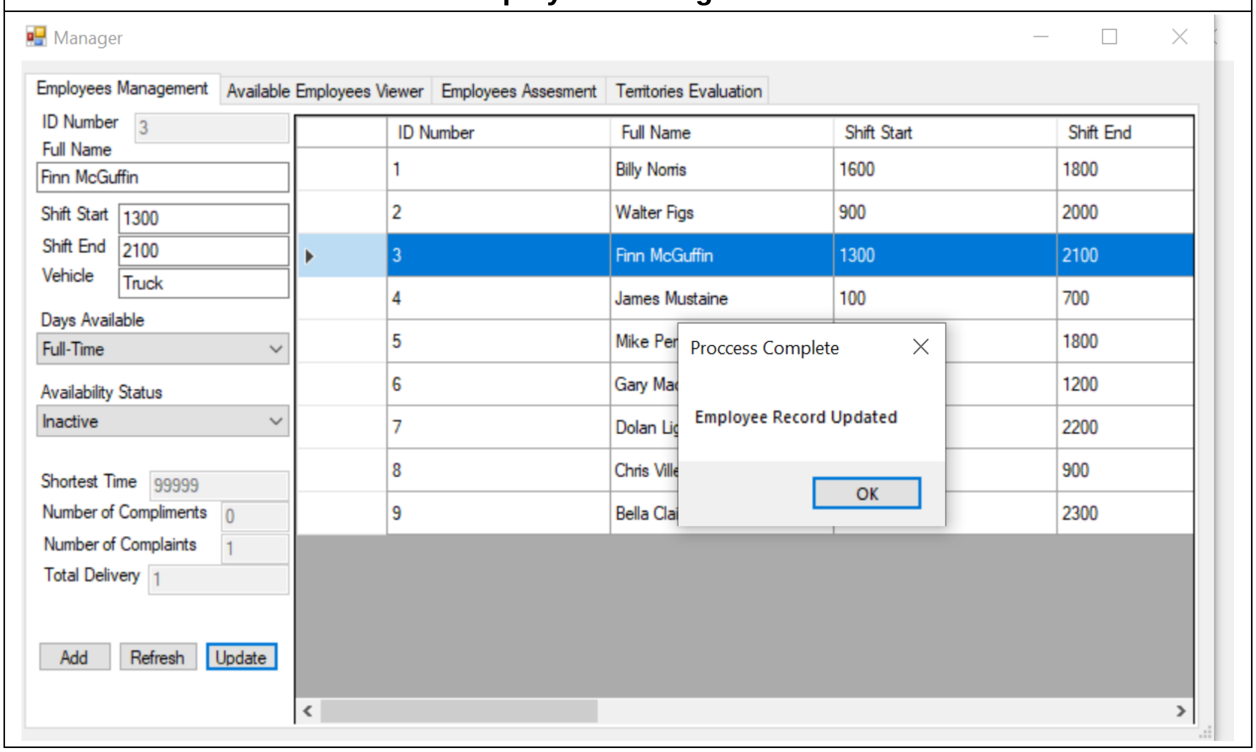
Select User



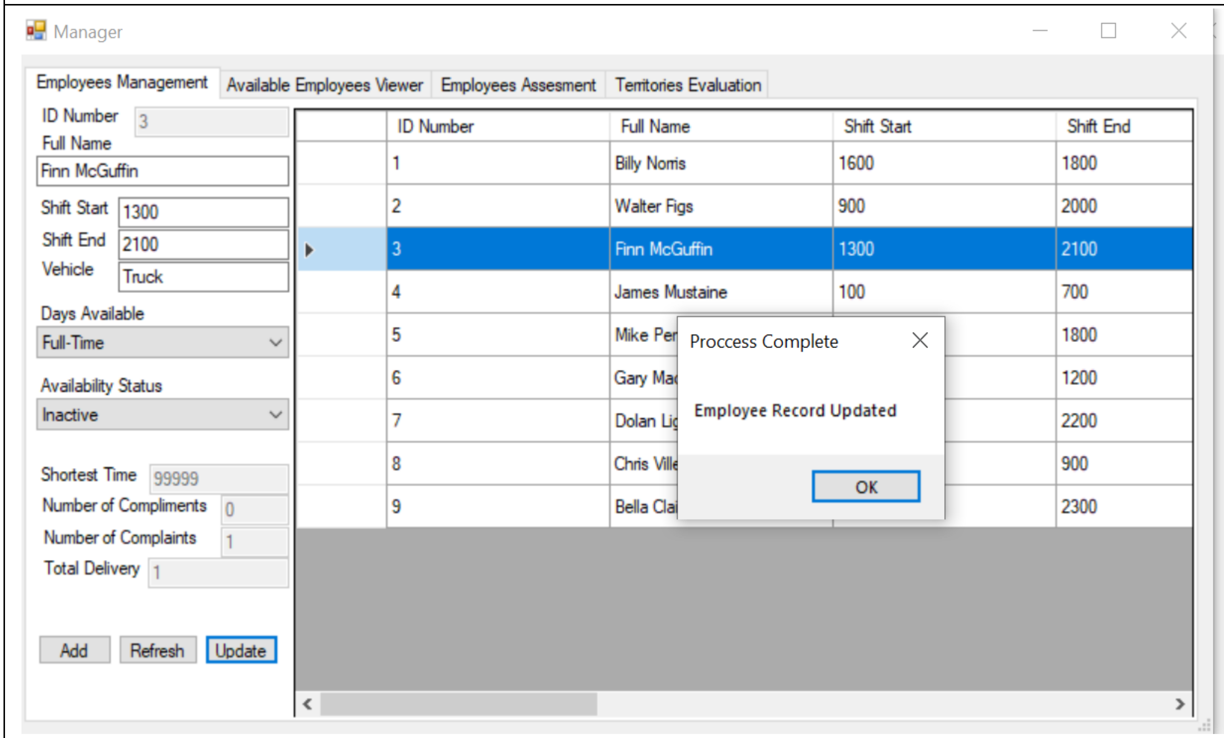
Path-Finder



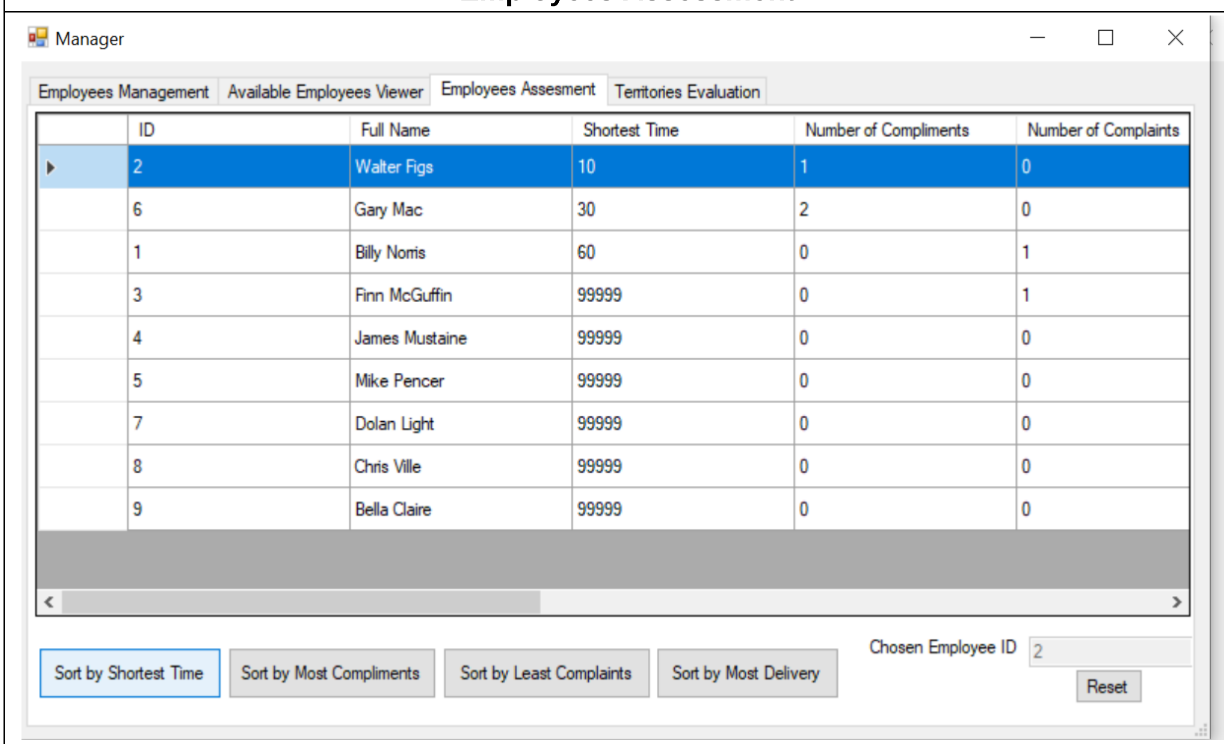
Employees Management



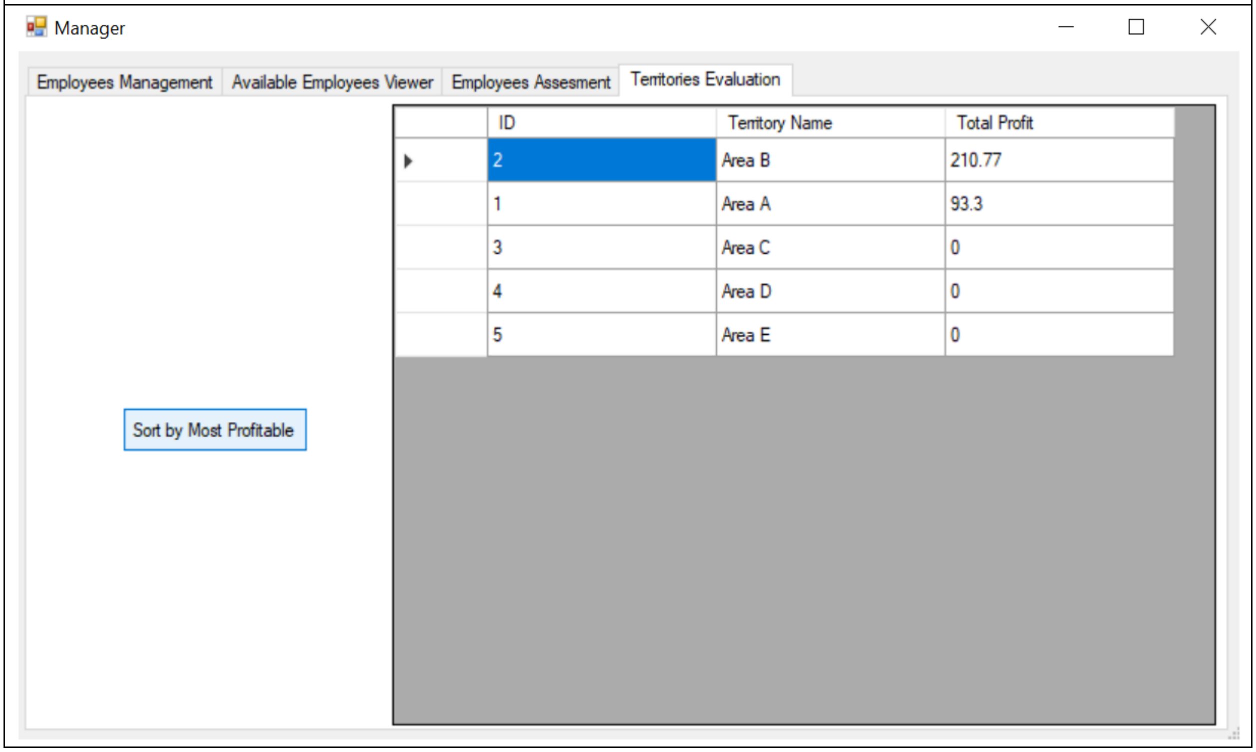
Available Employees Viewer



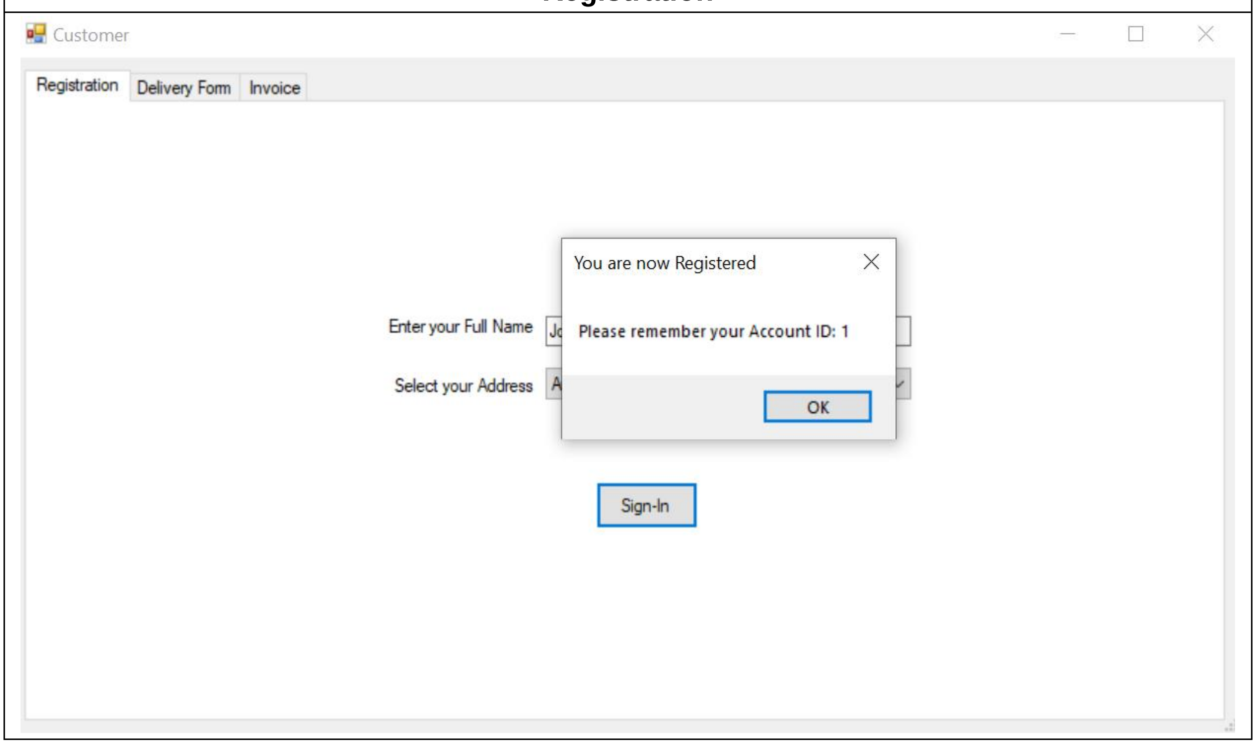
Employees Assessment



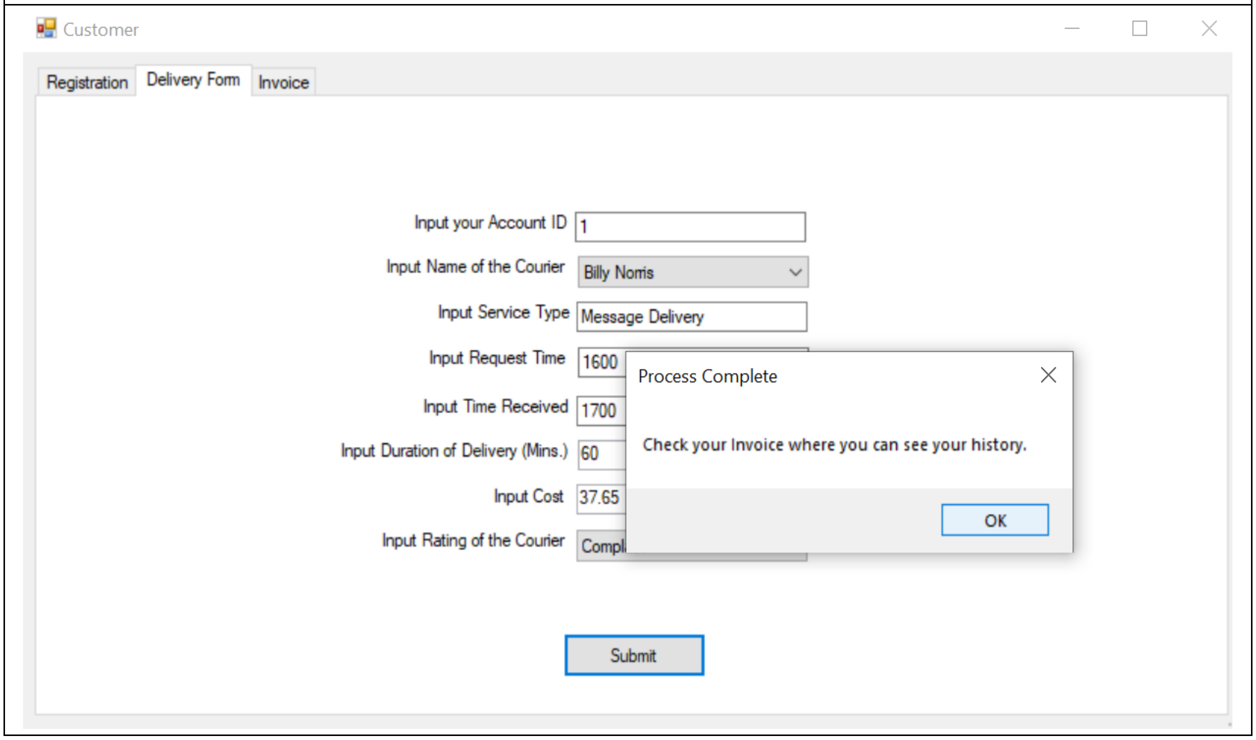
Territories Evaluation



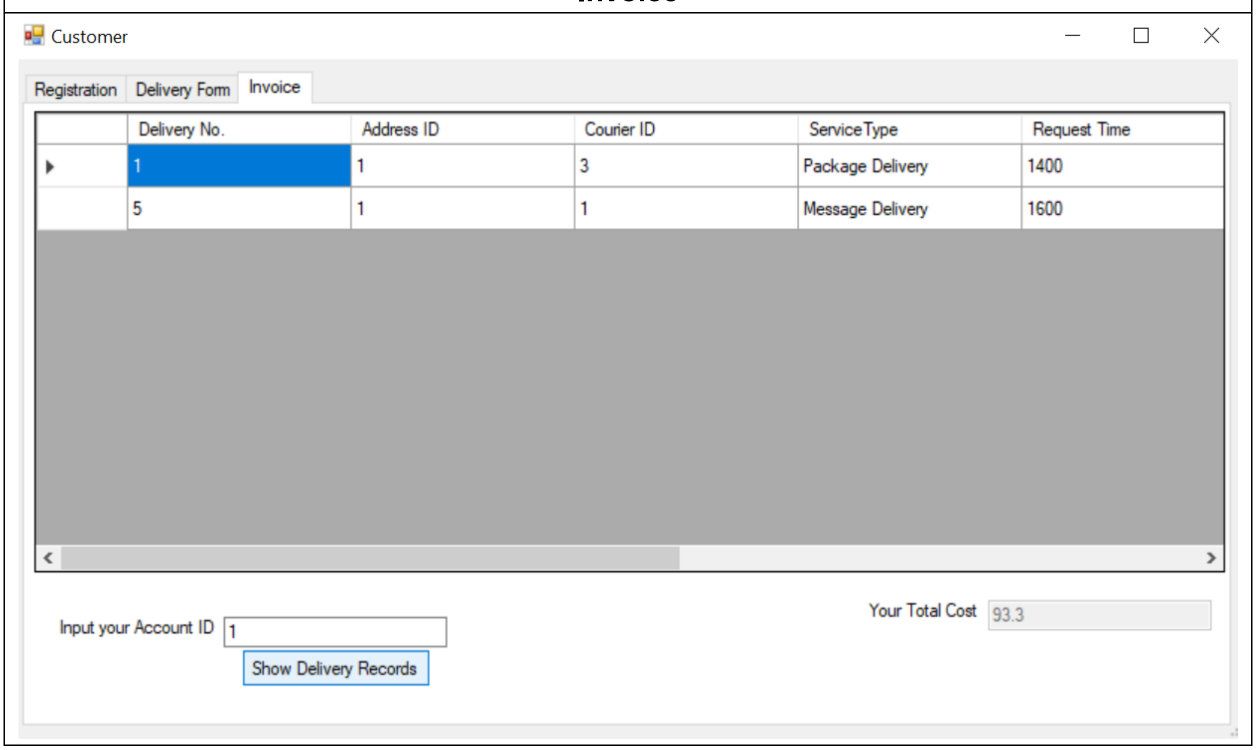
Registration



Delivery Form



Invoice



Gannt Chart

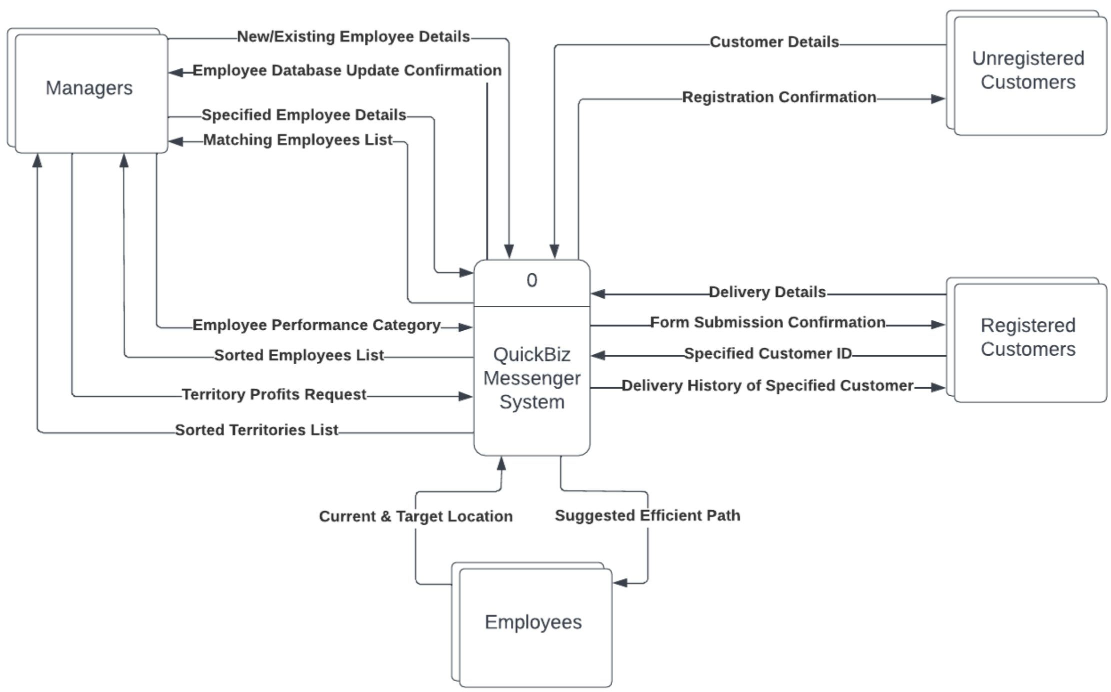
Duration	June			Indefinite
	Week 1	Week 2	Week 3	
Planning				
Design				
Development				
Testing				
Deployment				
Maintenance				

Work Breakdown Structure

Task No.	Description	Duration (Days)	Predecessor Task(s)
1	Planning	2	
2	Design	5	1
3	Development	7	2
4	Testing	3	3
5	Deployment	2	4
6	Maintenance	Undetermined	5

Present Data Flow Diagram

Context Diagram | QuickBiz Messenger



Level 0 Diagram | QuickBiz Messenger